



An experimental and kinetic modeling study on NH₃/air, NH₃/H₂/air, NH₃/CO/air, and NH₃/CH₄/air premixed laminar flames at elevated temperature

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ABSTRACT

Ammonia combustion helps to realize the off-site use of renewable energy. To understand the combustion characteristics of NH₃, laminar burning velocities (S_L) of NH₃/air, NH₃/H₂/air, NH₃/CO/air, and NH₃/CH₄/air with various mole fractions of H₂, CO, and CH₄ in fuel (α), equivalence ratios ($\phi = 0.7\sim 1.4$) at elevated initial temperature ($T_i = 298\sim 423$ K) were measured in a constant-volume combustion vessel. Based on updated chemistry of NH₂, NNH, N₂H₂, and NO_x, a detailed kinetic model with 169 species and 1268 elementary reactions was developed and validated against the S_L , NO_x emissions, NH₃/NO interaction, and ignition delay times measured by this work and literature. The experimental results show that S_L of NH₃/H₂/air, NH₃/CO/air, and NH₃/CH₄/air flames have different dependence on α . It is nearly exponential for NH₃/H₂/air flame, non-linear for NH₃/CO/air flame, and nearly linear for NH₃/CH₄/air flame. NH₃/air flame has the strongest temperature dependence, followed by NH₃/H₂/air, NH₃/CH₄/air, and NH₃/CO/air flames. The updated NH₃ chemistry, especially NNH chemistry (NNH = N₂ + H, N₂H₂ + H = NNH + H₂, and N₂H₂ + O = NNH + OH), enables the present model obtain better temperature dependence of NH₃/air flame. Kinetic modeling analysis shows that the production and reduction of NO_x are preferred by the lean and stoichiometric flames of NH₃/H₂/air, NH₃/CO/air, and NH₃/CH₄/air because of the increased concentrations of NH_i and H/O/OH radicals. In the lean flames of NH₃/H₂/air, NH₃/CO/air, and NH₃/CH₄/air, H₂, CO, and CH₄ can promote the conversion of NH₂ to NO, and relatively inhibit the NO reduction by NH₂, resulting in the increase in NO emissions. In the rich flames of NH₃/H₂/air and NH₃/CO/air, NH₂ and NH can produce N₂H₂ and N₂H₃ through reactions of NH₂ + NH = N₂H₂ + H and NH₂ + NH = N₂H₃, which dominate the promotion and inhibition of laminar flame propagation, respectively.

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1. Introduction

As a carbon-neutral fuel and renewable energy carrier, ammonia (NH₃) can help to reduce CO₂ emissions. Benefiting from the milder liquefaction conditions and lower storage and transportation costs than hydrogen (H₂), NH₃ has received more and more attention recently [1]. NH₃ synthesis by thermochemical or electrochemical methods can realize long-term storage and global

transportation of valley, surplus, and abandoned power production from renewable energy [2]. The transported NH₃ is directly burned instead of being cracked into H₂ for secondary utilization, which can improve the utilization efficiency of NH₃ in boilers, gas turbines, and internal combustion engines [3].

However, compared with traditional hydrocarbon fuels, NH₃ still faces problems such as lower flame speed, narrower flammability limits, and higher NO_x formation [2]. Currently, methods like blending with highly reactive fuels [4–7], increasing the initial temperature [4,8], and oxygen enrichment [4,9] have been adopted to improve the oxidation of NH₃. Among them, blending hydrocarbon fuels with a dual fuel burner and raising the initial

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temperature with an air preheater are the most effective and economical ways to improve the reactivity of NH_3 . Therefore, it is necessary to study NH_3 chemistry with and without the addition of highly reactive fuels at elevated temperatures.

Kinetic models help us better understand NH_3 chemistry and facilitate stable, low-emission combustion of NH_3 in industrial applications. The developed kinetic models can be validated using laminar burning velocity (S_L). Tian et al. [10] first developed a detailed kinetic model of NH_3/CH_4 combustion by identifying combustion intermediates and products. Okafor et al. [11] re-evaluated the NH_3 and CH_4 sub-mechanisms based on Tian kinetic model (simplified as the Tian model in the following) [10] and GRI 3.0 model [12], a reduced kinetic model for NH_3/CH_4 S_L measurement was developed. Meanwhile, Shrestha et al. [13] developed a detailed NH_3 -NO_x kinetic model, which was subsequently modified based on the measured S_L of $\text{NH}_3/\text{H}_2/\text{air}$ and NH_3/air mixtures with enriched oxygen content at elevated pressure and temperature [4]. Mei et al. separately developed two kinetic models of NH_3/air combustion with the oxygen enrichment [9] and $\text{NH}_3/\text{syngas}$ combustion at high pressure [7] by modifying and simplifying the kinetic model developed by Shrestha et al. [4]. Additionally, Han et al. [5] measured the S_L of NH_3/air , $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$, $\text{NH}_3/\text{CH}_4/\text{air}$, and $\text{NH}_3/\text{syngas}/\text{air}$ mixtures at ambient temperature and pressure. However, almost all of the above kinetic models are not successful to obtain satisfactory results in predicting the S_L . Recently, Stagni et al. [14] have developed a detailed kinetic model based on theoretical calculations and oxidation products measured in a jet-stirred reactor and a flow reactor. Afterward, Arunthanayothin et al. [15] also developed a detailed NH_3/CH_4 combustion model based on NH_3 chemistry from the Stagni model [14] and adding the CH_4 chemistry selected from Metcalfe et al. [16]. However, the absence of several important NH_2 combination reactions in the Stagni model [14] results in a large deviation in predicting S_L . For instance, the absence of $\text{NH}_2 + \text{NH} = \text{N}_2\text{H}_3$ and $\text{NH}_2 + \text{NH}_2 = \text{N}_2\text{H}_2 + \text{H}_2$ makes NH_2 easier to generate N_2H_2 through $\text{NH}_2 + \text{NH} = \text{N}_2\text{H}_2 + \text{H}$, leading to a rapid increase of S_L in rich conditions. Meanwhile, Zhang et al. [17] developed a new NH_3/H_2 kinetic model based on Li & Sarathy's [18] updates on NH_2 , N_2H_2 , and NNH chemistry, and validated the model by the measurement of $\text{NH}_3/\text{H}_2/\text{O}_2/\text{N}_2$ oxidation products in a jet-stirred reactor.

In general, the developed kinetic models based on oxidation products measurement have a relatively high NH_3 reactivity, resulting in an excessively high prediction of S_L , such as the Stagni model [14], Arunthanayothin model [15], Zhang model [17], and Tang model [19], etc. Moreover, the developed kinetic models based on the S_L measurement usually ignore the species that have a minor effect on S_L , making it difficult to predict the oxidation products, such as the Okafor model [11], etc. In addition, according to our previous work [8] and literature [4], current kinetic models also have insufficient temperature dependence in predicting S_L .

Therefore, this work measured the S_L of NH_3/air , $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$, and $\text{NH}_3/\text{CH}_4/\text{air}$ mixtures at elevated initial temperatures (T_i) with various equivalence ratios (ϕ). At one hand, the S_L database of NH_3/air , $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$, and $\text{NH}_3/\text{CH}_4/\text{air}$ mixtures is broadened. On the other hand, it serves as the experimental foundation for the kinetic model validation. In addition, to improve the flexibility in predicting the oxidation products of NH_3 with highly reactive fuels, a new detailed kinetic model is developed and verified against the wide experimental data for S_L , oxidation products, and ignition delay time (IDT). Kinetic analysis including sensitivity analysis and consumption flux analysis for laminar flame propagation characteristics of NH_3 is discussed.

2. Experimental and kinetic modeling details

2.1. Experimental details

The experiments were performed in a cylindrical stainless steel combustion vessel. The detailed description is shown in [8]. A pressure transmitter (Rosemount, 3051TG) is used for monitoring the chamber pressure with the accuracy of ± 0.04 F.S. All gas is not allowed to fill the chamber until the pressure of residual gas is pumped to < 0.006 bar by the vacuum pump. The temperature is monitored by a K-type thermocouple and controlled by PID controllers with a maximum of 0.5 K uncertainty at $T_i = 423$ K. To ensure a homogeneous mixture, a fan is turned on for a few seconds. After the intake is completed, two pulse electrodes with an ignition coil time charge of 1~2 ms are used for ignition. The spherically propagating flame is recorded by a high-speed camera (Phantom, VEO 640, 7000 fps) and a schlieren system. The nonlinear extrapolation method proposed by Kelly and Law [20] is used for S_L measurement. To avoid the ignition effect and wall confinement effect, the radius range of 20~45 mm is determined for estimating S_L . S_L measurement is repeated 2 to 3 times to obtain the average value and standard deviation.

All gas used in this work is provided by Tenglong Gas Co., Ltd. The purities of NH_3 , H_2 , CO , and CH_4 are all higher than 99.999%, and the purity of CO is higher than 99.99%. To reduce the adsorption of NH_3 on stainless steel, the vessel passivation method suggested in previous studies [7,8,21] was adopted. The stainless steel wall and CO have the potential to produce $\text{Fe}(\text{CO})_5$ through the reaction of $\text{Fe} + 5\text{CO} = \text{Fe}(\text{CO})_5$ under the high pressure conditions. It is a strong flame inhibitor [22]. Our experiment is carried out at normal pressure, therefore the influence of $\text{Fe}(\text{CO})_5$ can be ignored.

All experimental conditions investigated in this work are shown in Table 1. All experiments were performed at P_i (initial pressure) = 1 atm. α is defined as the mole fraction of $\text{H}_2/\text{CO}/\text{CH}_4$ in fuel. ϕ is determined considering the stoichiometry of overall reaction: $4\text{NH}_3 + 3\text{O}_2 = 2\text{N}_2 + 6\text{H}_2\text{O}$. Experimental uncertainties are mainly composed of radiation heat loss, non-linear extrapolation method, buoyancy effect, equivalence ratio, and measurement randomness, represented by error bars (0.8~5.8 cm/s) in the figures of this work. Detailed descriptions of the data processing method and evaluation method of experimental uncertainties are shown in S1~S2 of Supplementary Materials.

2.2. Kinetic modeling details

The C/H/O sub-mechanism in the present model is selected from the Aramco 3.0 model [23]. Good performances have been achieved in predicting the combustion of $\text{C}_0\sim\text{C}_2$ hydrocarbon fuels by Aramco 3.0 model [23]. Since CH_4 chemistry is dominated by the $\text{C}_0\sim\text{C}_2$ sub-mechanism, the $\text{C}_0\sim\text{C}_2$ sub-mechanism is used in the present model. The NH_3 sub-mechanism is selected from our previous study [8] and updated with recent stud-

Table 1
Experimental conditions in this work.

Fuel Type	Variables		
	α	T_i (K)	ϕ
NH_3	0	298, 360	0.8~1.3 ($\Delta=0.1$)
	0	423	0.8~1.4 ($\Delta=0.1$)
NH_3/H_2	0~0.7 ($\Delta=0.1$)	298	1.0
	0.3	298, 360, 423	0.7~1.4 ($\Delta=0.1$)
NH_3/CO	0~0.9 ($\Delta=0.1$)	298	1.0
	0.5	298, 360, 423	0.7~1.4 ($\Delta=0.1$)
NH_3/CH_4	0~1.0 ($\Delta=0.1$)	298	$\phi = 1.0$
	0.5	298, 360, 423	0.7~1.4 ($\Delta=0.1$)

Table 2
Updated important reactions of NH_3 chemistry.

No.	Reaction	Rate constants (cm, mole, s, cal)			Reference
		A	n	Ea	
R1	$\text{NH}_2 + \text{H} = \text{NH} + \text{H}_2$	1.09E5	2.590	1812.3	[18]
R2	$\text{NH}_2 + \text{OH} = \text{NH} + \text{H}_2\text{O}$	4.04E4	2.520	-616.0	[18]
R3	$\text{NH}_2 + \text{O} = \text{HNO} + \text{H}$	6.6E14	-0.5	0.0	[33]
R4	$\text{NH}_2 + \text{NO} = \text{N}_2 + \text{H}_2\text{O}$	2.60E19	-2.369	870.0	[35]
R5	$\text{NH}_2 + \text{NO} = \text{NNH} + \text{OH}$	4.30E10	0.294	-866.0	[35]
R6	$\text{NH}_2 + \text{HO}_2 = \text{NH}_3 + \text{O}_2$	6.04E18	-1.91	306.0	[24]
	DUP	5.91E7	1.59	-1373.0	[24]
R7	$\text{NH}_2 + \text{HO}_2 = \text{H}_2\text{NO} + \text{OH}$	2.19E9	0.791	-1428.0	[24]
R8	$\text{NH}_2 + \text{HO}_2 = \text{HNO} + \text{H}_2\text{O}$	1.02E12	0.166	-938.0	[24]
R9	$\text{NH} + \text{O} = \text{NO} + \text{H}$	9.20E13	0	0	[32]
R10	$\text{NNH} = \text{N}_2 + \text{H}$	7.68E07	1.730	4282.4	[18]
R11	$\text{N}_2\text{H}_2 + \text{H} = \text{NNH} + \text{H}_2$	4.82E08	1.76	739.2	[18]
R12	$\text{N}_2\text{H}_2 + \text{O} = \text{NNH} + \text{OH}$	1.11E8	1.62	804.8	[18]

ies [18,24]. The updated important reactions of NH_3 chemistry are shown in Table 2. Li and Sarathy [18] performed energy calculations through high level quantum chemical methods and obtained the slightly lower barrier heights of $\text{NH}_2 + \text{H} = \text{NH} + \text{H}_2$ (R1) and $\text{NH}_2 + \text{OH} = \text{NH} + \text{H}_2\text{O}$ (R2). Correspondingly, the reaction rate of R1 is faster than the Mackie et al. [25] and Linder et al. [26] by a factor of 2~4. Furthermore, the reaction rate of R2 within 800~2000 K is faster than the Klippenstein et al. [27], Dean and Bozzelli et al. [28], Mackie et al. [25], and Cohen et al. [29] by a factor of 1.5~3.0. A better agreement with the experimental values has been achieved. Therefore, the rate constants of R1 and R2 in the work are adopted. $\text{NH}_2 + \text{O} = \text{HNO} + \text{H}$ (R3) is the main pathway for NO production, while $\text{NH}_2 + \text{NO} = \text{N}_2 + \text{H}_2\text{O}$ (R4) and $\text{NH}_2 + \text{NO} = \text{NNH} + \text{OH}$ (R5) are the main pathway of NO reduction. Since a good performance has been achieved for the prediction of NO formation by the Glarborg model [30], the rate constants of R3~R5 in this work are consistent with the Glarborg model [30]. $\text{NH}_2 + \text{HO}_2 = \text{NH}_3 + \text{O}_2$, $\text{NH}_2 + \text{HO}_2 = \text{H}_2\text{NO} + \text{OH}$, and $\text{NH}_2 + \text{HO}_2 = \text{HNO} + \text{H}_2\text{O}$ (R6~8) are three elementary reactions with $\text{NH}_2 + \text{HO}_2$ as reactant. Klippenstein and Glarborg [24] recently updated the rate constants of R6~8, reducing the reactivity of NH_3 appropriately. The updated rate constants of R6~8 are also used in our work. Among the three reactions, R6 has the greatest influence on NH_3 oxidation. Chavarrio et al. [31] recently also attempted to update the rate constants of R6 by theoretical calculations, but their update severely lowered the NH_3 reactivity. Stagni et al. [14] evaluated the rate constants of R6 through theoretical calculations, but R8 was ignored in their evaluation, which inevitably increased the influence of R6 and R7 on NH_3 oxidation. Therefore, neither the evaluation of Chavarrio et al. [31] nor Stagni et al. [14] were adopted. In the lean condition, $\text{NH} + \text{O} = \text{NO} + \text{H}$ (R9) is the secondary pathway of NH consumption, and the main pathway of NH consumption is $\text{NH} + \text{NO} = \text{N}_2\text{O} + \text{H}$ [5]. A faster reaction rate of R9 would promote NO production and inhibit N_2O production. Cohen et al. [29] suggested that the pre-exponential factor (A) of R9 is $3.0\text{E}13$, while Baulch et al. [32] believed that it should be $9.2\text{E}13$. A recommended by Baulch et al. [32], which is validated against the N_2O emission in the literature [17], is adopted in this work. Since $\text{NNH} = \text{N}_2 + \text{H}$ (R10) can promote the conversion of NNH to N_2 and produce a large number of H radicals, it has a significant positive effect on S_L . Although A of R10 varies widely among the different kinetic models, with the lowest A being $6.500\text{E}07$ recommended by Miller et al. [33] and the highest A being $1.000\text{E}09$ suggested by Klippenstein et al. [27]. The updated rate constants of R10 evaluated by [18] can significantly improve the temperature dependence of NH_3 chemistry. Meanwhile, N_2H_2 is an important intermediate in the rich NH_3 chemistry, and the updated rate constants of $\text{N}_2\text{H}_2 + \text{H} = \text{NNH} + \text{H}_2$ (R11) and $\text{N}_2\text{H}_2 + \text{O} = \text{NNH} + \text{OH}$ (R12) [18] can significantly increase the

temperature dependence of NH_3 chemistry with the addition of H_2 . Therefore R10~12 are updated in this work. The C-N interaction mechanism was selected from the Glarborg model [30], and the HCN and HNCO sub-mechanisms were also updated according to [18]. The comparison of laminar burning velocity of NH_3 /air mixtures simulated by the original model and the present model is shown in S4 of Supplementary Materials.

The thermodynamic data of the relevant species in the C/H/O sub-mechanism were from Aramco 3.0 model [23], and the thermodynamic data of the relevant species in the NH_3 sub-mechanism and C-N interaction sub-mechanism were from Glarborg model [30]. Thermodynamic data for NH_2 , N_2H_2 , and NNH species were updated according to [18]. The simulated S_L is calculated by the Premixed Laminar Flame Speed module of Chemkin Pro-software [34]. For each case, up to 500 grid points are used for simulation calculations. CURV and GRAD are set to 0.05 to ensure grid independence. Thermal diffusion (soret effect) is considered. Additionally, mixture-averaged transport is adopted for higher calculation efficiency.

3. Results and discussion

3.1. Measured and simulated s_L of NH_3 /air, NH_3/H_2 /air, NH_3/CO /air, and NH_3/CH_4 /air mixtures

Fig. 1 shows the measured and simulated S_L of NH_3 /air mixtures with $\phi = 0.8\sim 1.3$ and $T_i = 298\sim 423$ K at atmospheric pressure. Literature data [9,36~41] are also shown. It is visible that

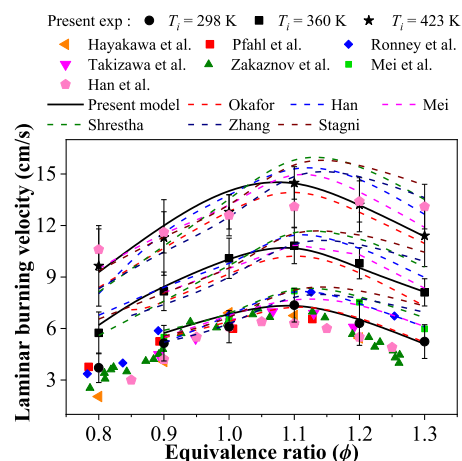


Fig. 1. Measured and simulated S_L of NH_3 /air mixtures with $\phi = 0.8\sim 1.3$, $T_i = 298\sim 423$ K, and $P = 1$ atm.

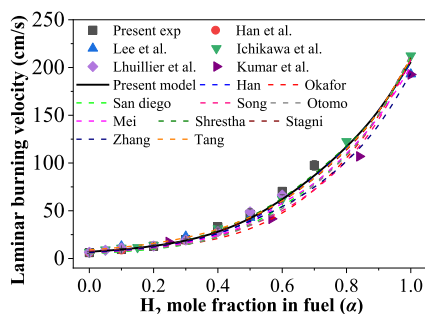


Fig. 2. Measured and simulated S_L of $\text{NH}_3/\text{H}_2/\text{air}$ with $\alpha = 0\sim 0.7$, $\phi = 1.0$, $T_i = 298$ K, and $P = 1$ atm.

the S_L measured by Mei et al. [9] and Ronney et al. [38] is the fastest, and the S_L measured by Han et al. [41] and Zakaznov et al. [40] is the slowest, while the S_L measured by this work is in between at $T_i = 298$ K. At $T_i = 423$ K, the S_L measured by us are comparable to those measured by Han et al. [41] in most cases. It also proves the correctness of the measurement in our experimental setup. Since the S_L simulated by the Tang model [19] is too fast, no comparison will be made in this work between the Tang model [19] and the present model. The NH_3 sub-mechanism in the Arunthanayothin model [15] is entirely derived from the Stagni model [14]. The S_L simulated by the Arunthanayothin model [15] is almost the same as those simulated by the Stagni model [14], therefore it is also excluded in Fig. 1. Good agreement can be observed at $T_i = 298\sim 423$ K between the measured and our simulated S_L while discrepancy is also presented at $T_i = 423$ K and $\phi = 1.0$. Generally, our model shows great improvement in predicting the S_L of NH_3/air rich flames due to the more complete N_2H_i sub-mechanism. The combination reaction ($\text{NH}_2 + \text{NH} = \text{N}_2\text{H}_2 + \text{H}$) and self-combination reaction ($\text{NH}_2 + \text{NH}_2 = \text{N}_2\text{H}_2 + \text{H}_2$) of NH_i can promote the production of active species (H_2) and active radical (H) to accelerate the S_L . If NH_2 and NH can be more converted to N_2H_3 by $\text{NH}_2 + \text{NH} = \text{N}_2\text{H}_3$, the production of N_2H_2 will be inhibited, resulting in the slowing down of S_L . Although a relatively complete N_2H_i sub-mechanism was also found by the Shrestha model [4], they doubled the rate constant of $\text{NH}_2 + \text{NH} = \text{N}_2\text{H}_2 + \text{H}$, leading to a large overestimation of rich flame. In the other models, the incomplete N_2H_i sub-mechanism causes significant overestimation.

Figs. 2 and 3 display the S_L of $\text{NH}_3/\text{H}_2/\text{air}$ mixtures with various α , ϕ , and T_i and literature data [5,42–45]. It can be seen from Fig. 2 that as α increases, the S_L of $\text{NH}_3/\text{H}_2/\text{air}$ mixtures increases exponentially. Our model can predict the S_L of $\text{NH}_3/\text{H}_2/\text{air}$ mixtures well over a wide range of H_2 mole fractions. The good agreement also has been achieved by the Han model [6], Mei model [7], Shrestha model [4], Stagni model [14], and Tang model [19], while obvious underestimation is observed by the San Diego model [46], Song model [35], Otomo model [47], Okafor model [11], and Zhang model [17] with the addition of medium NH_3 ratio. Similar phenomena are also found in Figs. 4 and 6. Wang et al. [48] believed that the underestimation is caused by the uncertainty of NH_3 chemistry. The net NH_3 burning rate reflects the uncertainty of NH_3 chemistry, which is proportional to the product of the NH_3 mole fractions in fuel and S_L . Fewer uncertainties of NH_3 chemistry are found with a low S_L and a low mole fraction of NH_3 in fuel. More uncertainties of NH_3 chemistry are found with moderate NH_3 mole fractions. In general, the uncertainty of NH_3 chemistry at room temperature has been reduced by the models developed recently.

Limited by the cellular instability [49], the S_L of $\text{NH}_3/\text{H}_2/\text{air}$ mixtures cannot be calculated with high H_2 mole fractions in

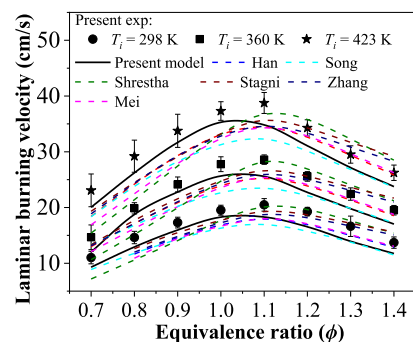


Fig. 3. Measured and simulated S_L of $\text{NH}_3/\text{H}_2/\text{air}$ mixtures with $\alpha = 0.3$, $\phi = 0.7\sim 1.4$, $T_i = 298\sim 423$ K, and $P = 1$ atm.

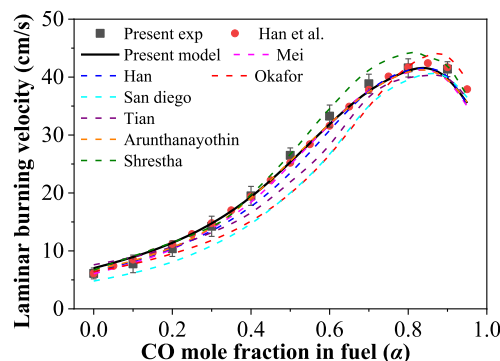


Fig. 4. Measured and simulated S_L of $\text{NH}_3/\text{CO}/\text{air}$ with $\alpha = 0\sim 0.9$, $\phi = 1.0$, $T_i = 298$ K, and $P = 1$ atm.

fuel at increased T_i in the lean flame. Therefore, S_L with the H_2 mole fraction of 30% at various ϕ and T_i was measured. In order not to affect the plotting, the S_L simulated by the models that deviate too much from the measured S_L will not be shown in Fig. 3. Good agreement can be observed in the lean and stoichiometric conditions, while underestimation in the rich conditions at $T_i = 298\sim 423$ K is also found. The little deviation between the measured S_L and the S_L simulated by the present model can be observed in Fig. 1. However, limited by the measurement technology and application scenarios, S_L of H_2/air flame at elevated T_i has hardly been found, leading to the possible insufficient temperature dependence of H_2 chemistry. At the same time, the large amount of H_2 produced by the NH_3 flame in rich conditions enlarges the underestimation of $\text{NH}_3/\text{H}_2/\text{air}$ rich flame in Fig. 3. The S_L simulated by other models is generally closer to the measured S_L in the rich conditions but deviates greatly from the measured S_L in the lean conditions, which is similar to the situation in Fig. 1, and the detailed analysis has been given above.

Figs. 4 and 5 present the S_L of $\text{NH}_3/\text{CO}/\text{air}$ mixtures with various α , ϕ , and T_i . As shown in Fig. 4, the S_L of $\text{NH}_3/\text{CO}/\text{air}$ mixtures increases first and then decreases with the increasing α , and reaches the maximum value of 41.6 cm/s at $\alpha = 0.85$. Since dry CO chemistry is dominated by $\text{CO} + \text{O}_2 = \text{CO}_2 + \text{O}$ and $\text{CO} + \text{O} + \text{M} = \text{CO}_2 + \text{M}$ without the participation of H radicals, the S_L of CO/air mixtures is close to zero [50]. As the mole fraction of NH_3 increases, the production of OH radicals gradually increases. The rapid oxidation of CO can be achieved through $\text{OH} + \text{CO} = \text{H} + \text{CO}_2$, resulting in the sharp increase of S_L . However, a large amount of NH_3 will reduce the adiabatic flame temperature, thereby reducing the S_L [48]. Generally, the S_L measured by the work is quite close to that measured by Han et al. [5]. Good agreement with measured S_L has been achieved by the present model, Han model [6], Mei model [7], and Arunthanayothin model

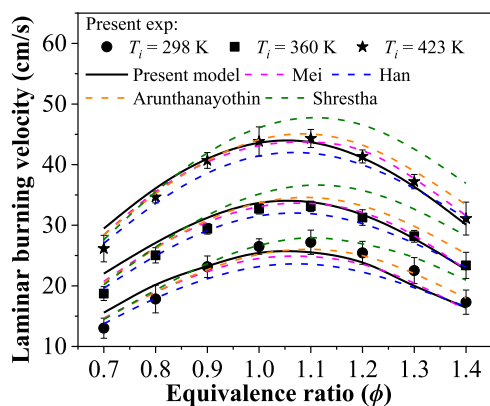


Fig. 5. Measured and simulated S_L of $\text{NH}_3/\text{CO}/\text{air}$ mixtures with $\alpha = 0.5$, $\phi = 0.7\sim 1.4$, $T_i = 298\sim 423$ K, and $P = 1$ atm.

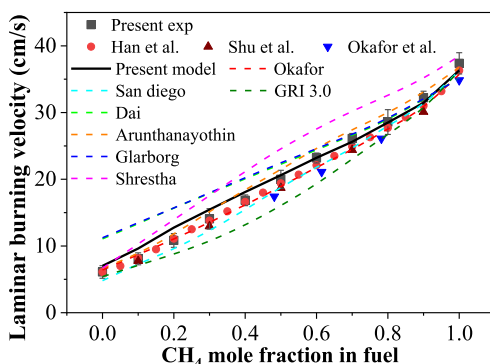


Fig. 6. Measured and simulated S_L of $\text{NH}_3/\text{CH}_4/\text{air}$ mixtures with $\alpha = 0\sim 1.0$, $\phi = 1.0$, $T_i = 298$ K, and $P = 1$ atm.

[15] for various α . Obvious underestimation by the other models is also observed, and the S_L simulated by these models will not appear in Fig. 5.

It can be seen from Fig. 5 that the present model can well reproduce the S_L of $\text{NH}_3/\text{CO}/\text{air}$ flame at a wide range of ϕ and T_i . A slight underestimation of the measured S_L is found by the Han model [6], but good simulation performances of the Mei model [7] and Arunthanayothin model [15] are also observed, while the Shrestha model [4] significantly overestimates the S_L in rich conditions. Meanwhile, the increasing temperature dependence of the Mei model [7], Arunthanayothin model [15], and Shrestha model [4] can be found. According to the simulation performances of the three models in Fig. 1, we can believe that it is caused by the overestimation of rich NH_3 chemistry.

Figs. 6 and 7 show the measured and simulated S_L of $\text{NH}_3/\text{CH}_4/\text{air}$ mixtures with various α , ϕ , and T_i . The S_L of $\text{NH}_3/\text{CH}_4/\text{air}$ mixtures with various α recorded by literature [5,11,51] is also shown in Fig. 6. The S_L measured by this work is quite close to that measured by Han et al. [5]. Except for the S_L of CH_4/air mixtures, which increases linearly with the increasing α . At $T_i = 298$ K and $\phi = 1.0$, the best agreement with the measured S_L has been obtained by the present model and Okafor model [11] among all the kinetic models. The changing trend of the measured S_L is also captured by the San Diego model [46] and Arunthanayothin model [15] well. Meanwhile, GRI 3.0 model [12] underestimates the S_L by approximately 25% with a medium mole fraction of NH_3 in fuel. The S_L simulated by the Dai model [52] and Glarborg model [30] have obvious overestimation with a high mole fraction of NH_3 in fuel. The Shrestha model [4] overestimates the NH_3 chemistry at almost every α . Therefore, the S_L simulated by

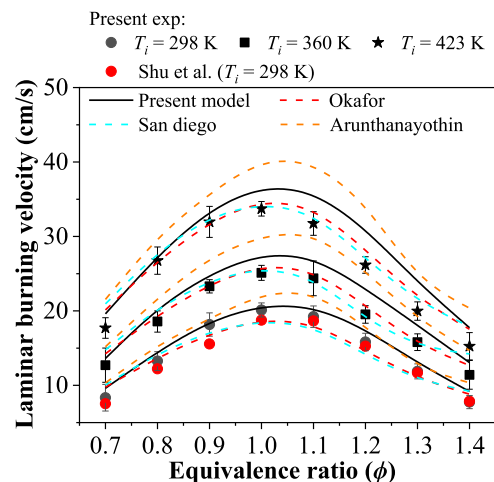


Fig. 7. Measured and simulated S_L of $\text{NH}_3/\text{CH}_4/\text{air}$ mixtures with $\alpha = 0.5$, $\phi = 0.7\sim 1.4$, $T_i = 298\sim 423$ K, and $P = 1$ atm.

the Dai model [52], Glarborg model [30], GRI 3.0 model [12], and Shrestha model [4] will not be shown in Fig. 7.

It is evident in Fig. 7 that the present model can predict the measured S_L well at $T_i = 298$ K, but there is a certain deviation at elevated temperatures as a result of the overestimated hydrocarbon chemistry. It usually favors the prediction of NH_3/CH_4 oxidation products, but not the prediction of S_L . Meanwhile, the Okafor model [11] and San Diego model [46] provide the best agreement with the measurement at elevated temperatures, while the Arunthanayothin model [15] presents the worst evaluation.

3.2. Model validation

To test the validity of the present model, several typical conditions in the literature are selected for validation. In addition to the S_L measured at room and elevated temperatures, the S_L measured at elevated pressure also needs to be verified. Fig. 8 shows the S_L of $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$, and $\text{NH}_3/\text{syngas}/\text{air}$ mixtures recorded by Wang et al. [48] and predicted by the present model as a function of ϕ at elevated pressures. The underestimation is observed with the addition of H_2 in stoichiometric conditions, but the S_L of $\text{NH}_3/\text{CO}/\text{air}$ mixtures at various ϕ and elevated pressures is reproduced by the present model well. Fig. 9 shows the good performance of the present model in simulating the S_L of $\text{NH}_3/\text{CH}_4/\text{air}$ mixtures at elevated pressures measured by Okafor et al. [53]. Fig. 10 shows the validation of the present model against the oxidation products of $\text{NH}_3/\text{H}_2/\text{O}_2/\text{N}_2$ mixtures as a function of temperature at $\phi = 0.25$ measured by Zhang et al. [17] in a jet-stirred reactor. The present model can predict the temperature of NH_3 reactivity onset, as well as the production of NO and N_2O , over a wide range of fuel compositions. Fig. 11 shows the concentrations of NH_3 and NO measured by Stagni et al. [14] and predicted by the present model as a function of temperature and oxygen concentrations. The present model can also reproduce the temperature of NH_3 reactivity onset and the NO formation under the conditions of oxygen enrichment and low temperature. Fig. 12 presents the ignition delay times of $\text{NH}_3/\text{O}_2/99\text{Ar}$ in a shock tube measured by Mathieu and Petersen [21] and simulated by the present model. It can be seen from the figure that the best simulation accuracy is achieved under high-pressure conditions, and the deviation is only found under conditions of low pressure and lean flame.

Additionally, S3 of Supplementary Materials shows the validation of the present model against the S_L , NO_x emissions, NH_3/NO interaction, and ignition delay time measured by other literature [4,6,7,14,15,17,21,48,53–59]. It is worth mentioning that the N_2O

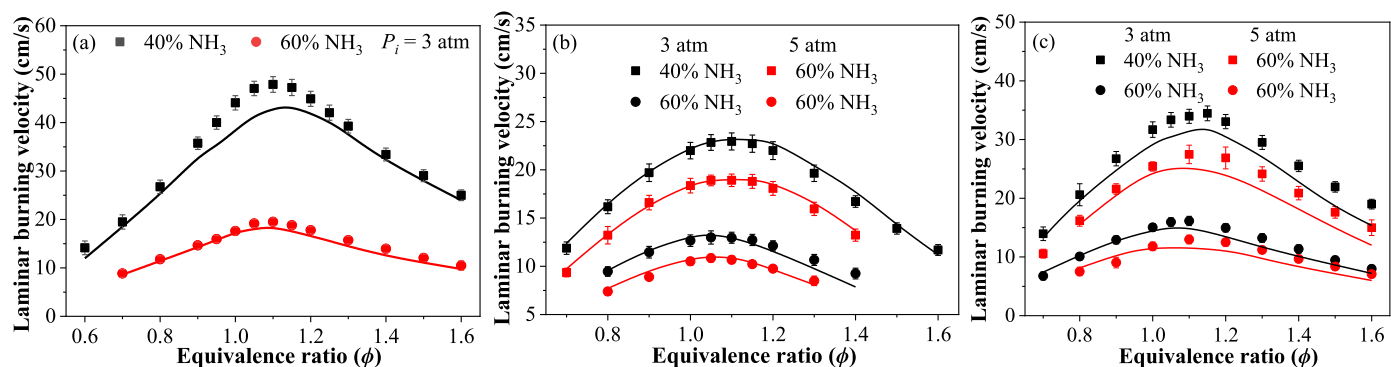


Fig. 8. Measured and simulated S_L of (a) $\text{NH}_3/\text{H}_2/\text{air}$, (b) $\text{NH}_3/\text{CO}/\text{air}$, and (c) $\text{NH}_3/\text{syngas}/\text{air}$ mixtures with various ϕ at elevated pressure ($T_i = 298$ K, Experimental data are from Wang et al. [48]).

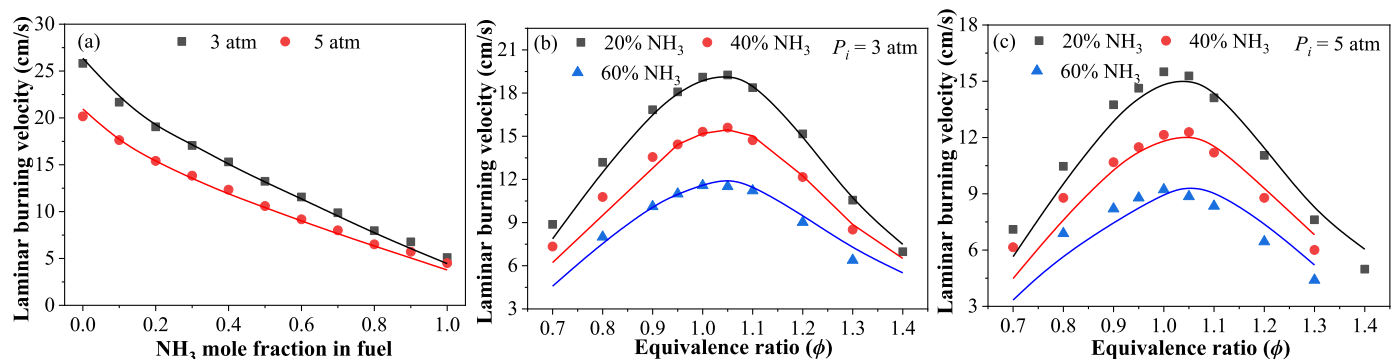


Fig. 9. Measured and simulated S_L of $\text{NH}_3/\text{CH}_4/\text{air}$ mixtures with various NH_3 mole fractions in fuel and ϕ at $P_i = 3$ and 5 atm ($T_i = 298$ K, experimental data are from Okafor et al. [53]).

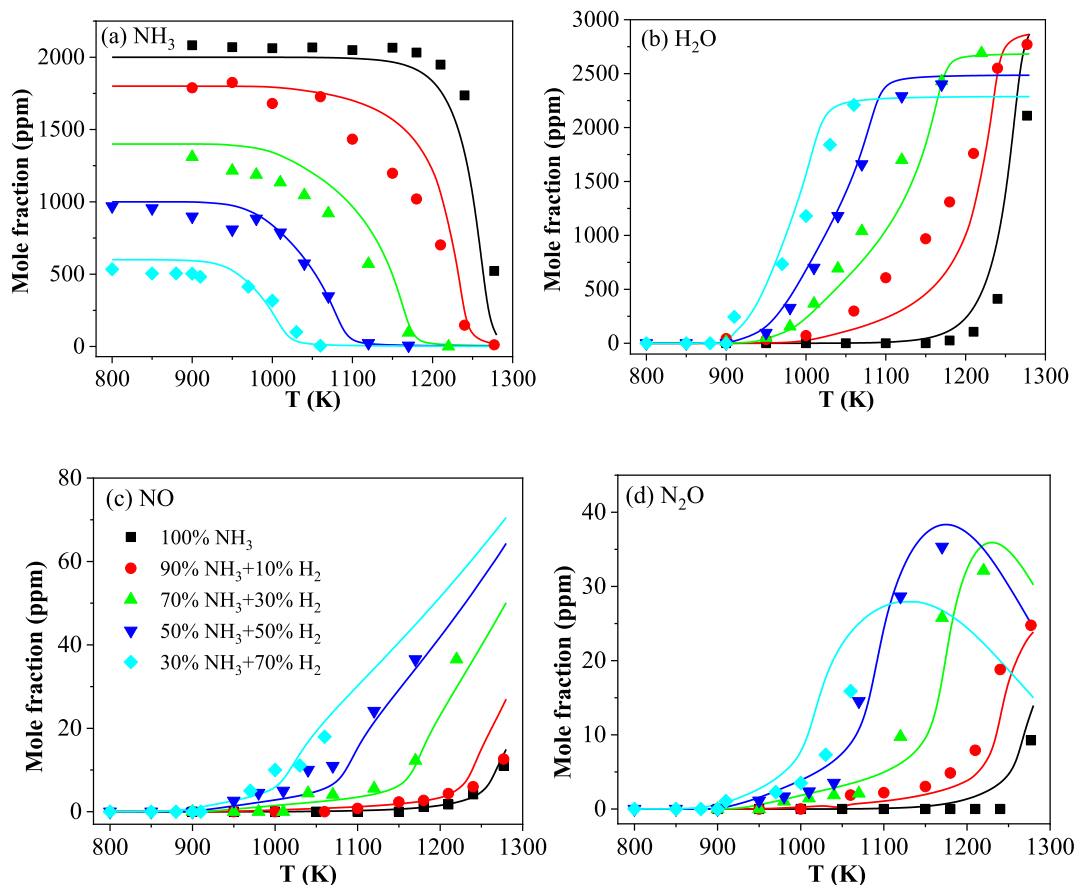


Fig. 10. Measured and simulated oxidation products of $\text{NH}_3/\text{H}_2/\text{O}_2/\text{N}_2$ mixtures as a function of temperature at $\phi = 0.25$ in a jet-stirred reactor ($\tau = 1$ s, experimental data are from Zhang et al. [17]).

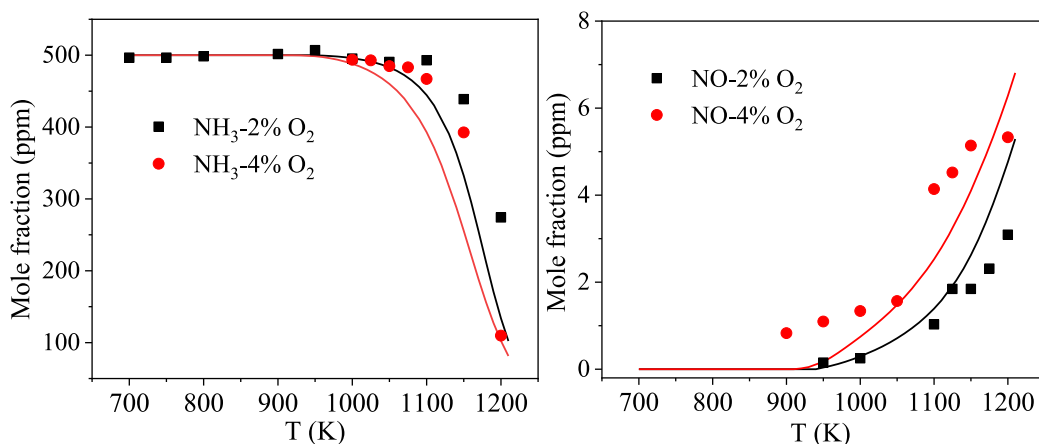


Fig. 11. Oxidation of 500 ppm of NH_3 . Measured and simulated concentrations of NH_3 and NO as a function of temperature and oxygen concentrations in a jet-stirred reactor ($\tau = 1.5$ s, experimental data are from Stagni et al. [14]).

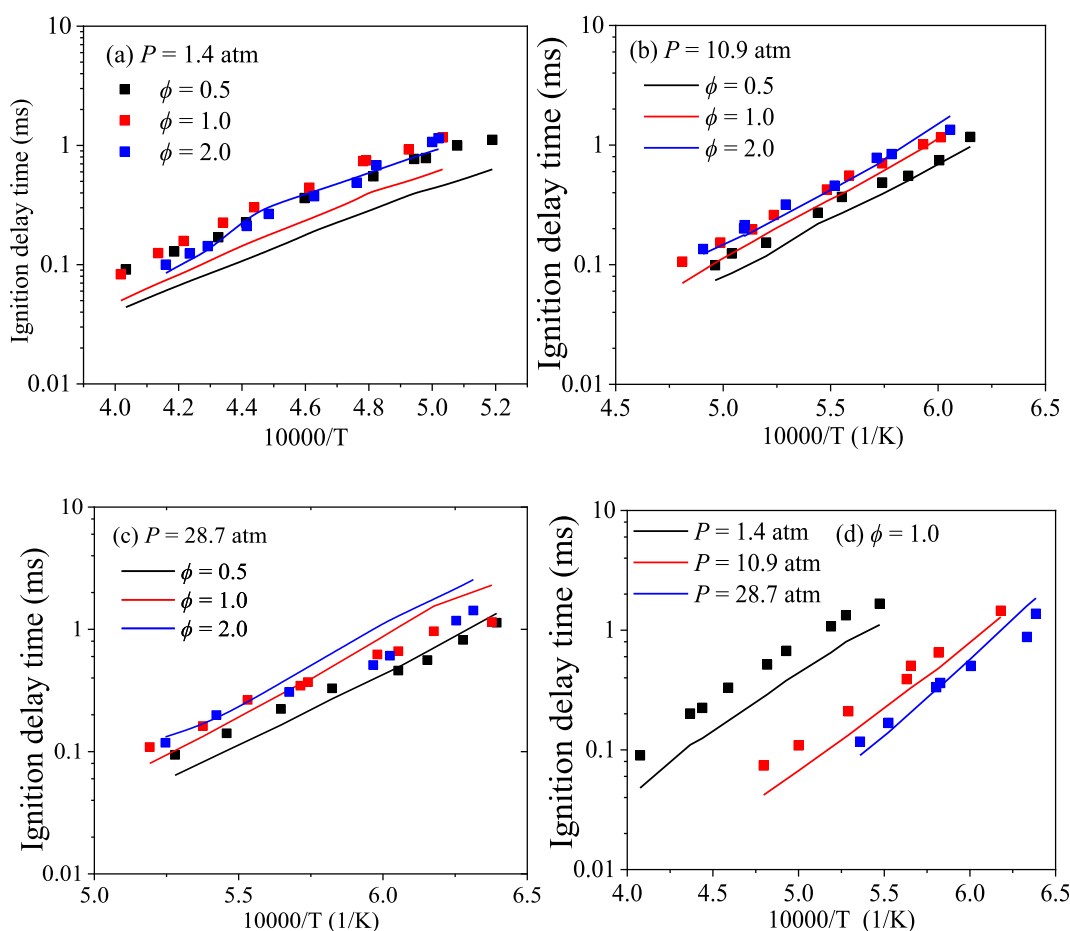


Fig. 12. Measured and simulated ignition delay times of $\text{NH}_3/\text{O}_2/99\text{Ar}$ in a shock tube (Experimental data are from Mathieu and Petersen [21]).

sub-mechanism in most kinetic models is significantly overestimated, as shown in S5 of *Supplementary Materials*. In general, only the present model, Zhang model [17], and Tang model [19] can predict the production trend of N_2O well.

3.3. Kinetic analysis

To understand the effects of fuel composition (α), equivalence ratio (ϕ), and initial temperature (T_i) on the laminar flame propagation characteristics, a detailed kinetic analysis based on this model was carried out.

3.3.1. Effect of fuel composition (α)

Fig. 13 shows the peak mole fractions of key radicals and adiabatic flame temperatures (T_{ad}) as a function of α in the $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$, and $\text{NH}_3/\text{CH}_4/\text{air}$ flames. It can be easily found from Figs. 2, 4, 6, and 13a that the variation trend of H radicals with α is very similar to that of S_L , implying that the chemical effect plays an important role in promoting the laminar flame propagation. The increase of S_L is usually caused by the chemical effect, thermal effect, and transport effect. According to our previous study [8] and literature [48], the promoted laminar flame propagation of NH_3 with the addition of highly reactive

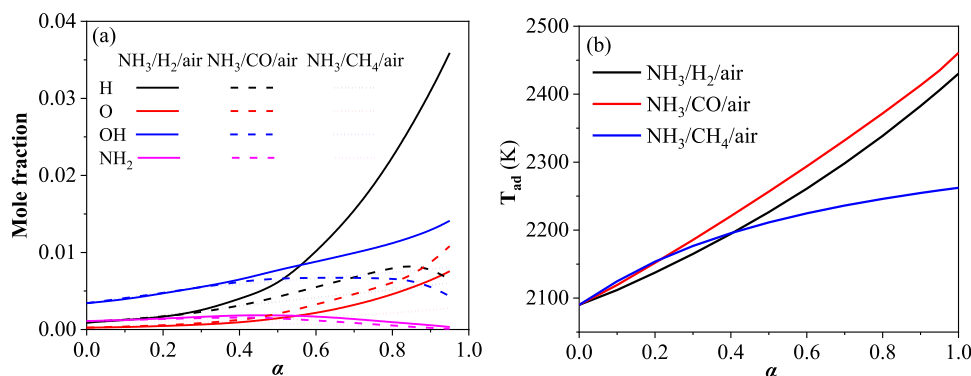


Fig. 13. (a) Peak mole fractions of key radicals and (b) adiabatic flame temperatures (T_{ad}) as a function of α in the $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$, and $\text{NH}_3/\text{CH}_4/\text{air}$ flame.

fuels ($\text{H}_2/\text{CO}/\text{syngas}$) is almost dominated by the chemical effect. NH_2 radicals increase first and decrease in all three flames. With the increase of α , the increased mole fractions of $\text{H}_2/\text{CO}/\text{CH}_4$ can promote the production of $\text{H}/\text{O}/\text{OH}$ active radicals, and then promote the oxidation of NH_3 , thereby increasing the mole fractions of NH_2 . Meanwhile, the lower mole fractions of NH_3 will make the mole fractions of NH_2 gradually decrease. It should be noted that the peak mole fractions of NH_2 appear at $\alpha = 0.4\sim 0.5$ in the $\text{NH}_3/\text{H}_2/\text{air}$ and $\text{NH}_3/\text{CO}/\text{air}$ flames, while the peak mole fraction of NH_2 appears at $\alpha = 0.1\sim 0.2$ in the NH_3/CH_4 flame, which is far earlier than $\text{NH}_3/\text{H}_2/\text{air}$ and $\text{NH}_3/\text{CO}/\text{air}$ flames. On one hand, the CH_4 in the NH_3/CH_4 flame has a lower temperature of reactivity onset than NH_3 in the NH_3/CH_4 flame [15]. CH_4 can achieve more advantages against the interaction with active radicals than NH_3 . Moreover, CH_4 oxidation is far more complicated than CO and H_2 oxidation. In the oxidation process of $\text{CH}_4 \rightarrow \text{CH}_3 \rightarrow \text{CH}_3\text{O} \rightarrow \text{CH}_2\text{O} \rightarrow \text{CHO} (\text{HCO}) \rightarrow \text{CO} \rightarrow \text{CO}_2$ [16], CH_4 and its intermediates will occupy more active radicals, resulting in the inhibition of NH_2 production. On the other hand, CH_3 and NH_2 will be converted into HCN through the C-N interaction pathways of $\text{CH}_3 + \text{NH}_2 \rightarrow \text{CH}_3\text{NH}_2 \rightarrow \text{CH}_2\text{NH}_2 \rightarrow \text{CH}_2\text{NH} \rightarrow \text{H}_2\text{CN} (\text{HCNH}) \rightarrow \text{HCN}$, in this case, the consumption of NH_2 is promoted [60]. According to [7], the thermal effects of H_2 and CO also have a positive influence on promoting the laminar flame propagation. In Fig. 13b, the thermal effect of CH_4 has a more positive influence on the laminar flame propagation than H_2 and CO at $\alpha = 0\sim 0.2$, but with the increase of α , the positive thermal effects of H_2 and CO on laminar flame propagation rapidly surpass that of CH_4 . A linear trend of $\text{NH}_3/\text{H}_2/\text{air}$ flame and a slowing trend of $\text{NH}_3/\text{CH}_4/\text{air}$ flame indicate that the thermal effect plays a decreasingly important role in the laminar flame propagation of $\text{NH}_3/\text{H}_2/\text{air}$ and $\text{NH}_3/\text{CH}_4/\text{air}$ mixtures, while the linear trend of $\text{NH}_3/\text{CO}/\text{air}$ flame demonstrates that the thermal effect plays an increasingly important role in the laminar flame propagation of $\text{NH}_3/\text{CO}/\text{air}$ mixtures at $\alpha > 0.85$.

As mentioned above there is a good correlation between the peak mole fraction of H radicals and S_L , 10 elementary reactions most sensitive to S_L at the flame position with the peak mole fraction of H radicals were selected for sensitivity analysis. The sensitivity coefficients of all elementary reactions are normalized by the sensitivity coefficients of $\text{H} + \text{O}_2 = \text{O} + \text{OH}$, and the elementary reactions related to N chemistry are marked in red.

Fig. 14 shows that the normalized sensitivity coefficients for S_L of NH_3/air , $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$, and $\text{NH}_3/\text{CH}_4/\text{air}$ mixtures at $\alpha = 0.5$, $\phi = 1.0$, and $T_i = 298$ K using the present model. In the NH_3/air flame, $\text{NH}_2 + \text{NH} = \text{N}_2\text{H}_2 + \text{H}$ has the largest positive sensitivity coefficient since H radicals can be produced by the reaction itself and the pathway of $\text{N}_2\text{H}_2 \rightarrow \text{NNH} \rightarrow \text{N}_2$. $\text{H} + \text{OH} + \text{M} = \text{H}_2\text{O} + \text{M}$ and $\text{H} + \text{O}_2 (+\text{M}) = \text{HO}_2 (+\text{M})$ have the largest negative sensitivity coefficients due to the consumption of

H radicals. In the $\text{NH}_3/\text{H}_2/\text{air}$ flame, $\text{NH}_2 + \text{NH} = \text{N}_2\text{H}_2 + \text{H}$ still has the most positive influence on S_L . In addition, as the result of the increased OH radicals in the radical pool with the addition of H_2 , the NH_3 oxidation is promoted by $\text{NH}_3 + \text{OH} = \text{NH}_2 + \text{H}_2\text{O}$, which also has an important and positive effect on S_L . It should be noted that $\text{NH}_2 + \text{NH} = \text{N}_2\text{H}_3$ has a great negative effect on the production of H radicals. If the combination reaction of NH_2 and NH produces N_2H_3 instead of N_2H_2 , the production of H radicals will be inhibited due to the completely different reaction pathway of N_2H_3 as shown in Fig. 15. In the $\text{NH}_3/\text{CO}/\text{air}$ flame, the high sensitivity coefficient of $\text{CO} + \text{OH} = \text{CO}_2 + \text{H}$ also implies the important positive effect of CO chemistry on S_L . In the $\text{NH}_3/\text{CH}_4/\text{air}$ flame, it is visible that the influence of NH_3 chemistry on S_L is significantly inhibited, such as $\text{NH}_2 + \text{NH} = \text{N}_2\text{H}_2 + \text{H}$. Due to the strong competition of CH_4 for active radicals, more active radicals will react with CH_4 to complete its oxidation process. Since $\text{NH}_3 + \text{OH} = \text{NH}_2 + \text{H}_2\text{O}$ is the main source of NH_2 , its importance will increase accordingly. Additionally, $\text{HCO} + \text{M} = \text{H} + \text{CO} + \text{M}$ has the largest positive sensitivity coefficient, indicating that the accelerated laminar flame propagation of $\text{NH}_3/\text{CH}_4/\text{air}$ mixtures is almost dominated by CO chemistry.

To understand the effect of different fuel blends on NH_3 chemistry, consumption flux analysis was performed in Fig. 15. For each reaction pathway, the consumption flux of specific species through an individual elementary reaction is calculated by the distance integration of reaction rate over the entire reaction zone [61], thus the unit of consumption flux is mole/cm²·s. The percentages represent the consumption flux of specific reaction pathways normalized by the consumption flux of $\text{NH}_3 + \text{OH} = \text{NH}_2 + \text{H}_2\text{O}$. Considering the weak correlation between the C-N interaction sub-mechanism and S_L [5], the reaction pathways containing the C-N interaction species will not be shown in Fig. 15d.

As shown in Fig. 15, the oxidation pathways of NH_3 mainly include the following: (1) The H-abstraction process of NH_3 : $\text{NH}_3 \rightarrow \text{NH}_2 \rightarrow \text{NH} \rightarrow \text{N} \rightarrow \text{N}_2$. (2) The oxidation process of NH_3 : $\text{NH}_3 \rightarrow \text{NH}_2 \rightarrow \text{HNO} \rightarrow \text{NO} (\rightarrow \text{N}_2\text{O}) \rightarrow \text{N}_2$. (3) The combination process of NH_3 : $\text{NH}_3 \rightarrow \text{NH}_2 \rightarrow \text{N}_2\text{H}_2 \rightarrow \text{NNH} \rightarrow \text{N}_2$. With the addition of $\text{H}_2/\text{CO}/\text{CH}_4$, the oxidation pathways of NH_3 mainly change as follows: (1) The importance of H radicals has been significantly improved. As can be seen from Fig. 13 above, the increase of H radical concentrations in the $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$, and $\text{NH}_3/\text{CH}_4/\text{air}$ flames is significantly higher than that of OH radical and O radical concentrations at $\alpha = 0.5$, indicating that $\text{H}_2/\text{CO}/\text{CH}_4$ significantly promote the oxidation of NH_3 and the production of H radicals. The higher collision probability between NH_i radicals and H radicals improves the importance of H radicals. (2) The combination reaction pathway of NH_i is inhibited. The reactions between NH_i radicals and other active radicals ($\text{H}/\text{O}/\text{OH}$) are promoted and the collision probability of NH_i radicals also reduces

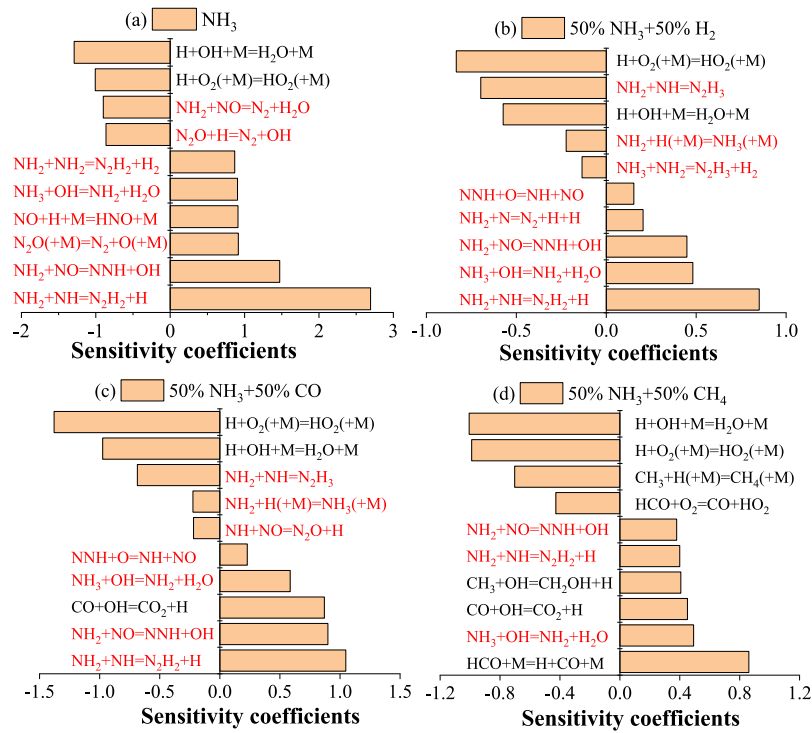


Fig. 14. Normalized sensitivity coefficients for S_L of (a) NH_3/air , (b) $\text{NH}_3/\text{H}_2/\text{air}$, (c) $\text{NH}_3/\text{CO}/\text{air}$, and (d) $\text{NH}_3/\text{CH}_4/\text{air}$ mixtures at $\alpha = 0.5$, $\phi = 1.0$, and $T_i = 298$ K using the present model.

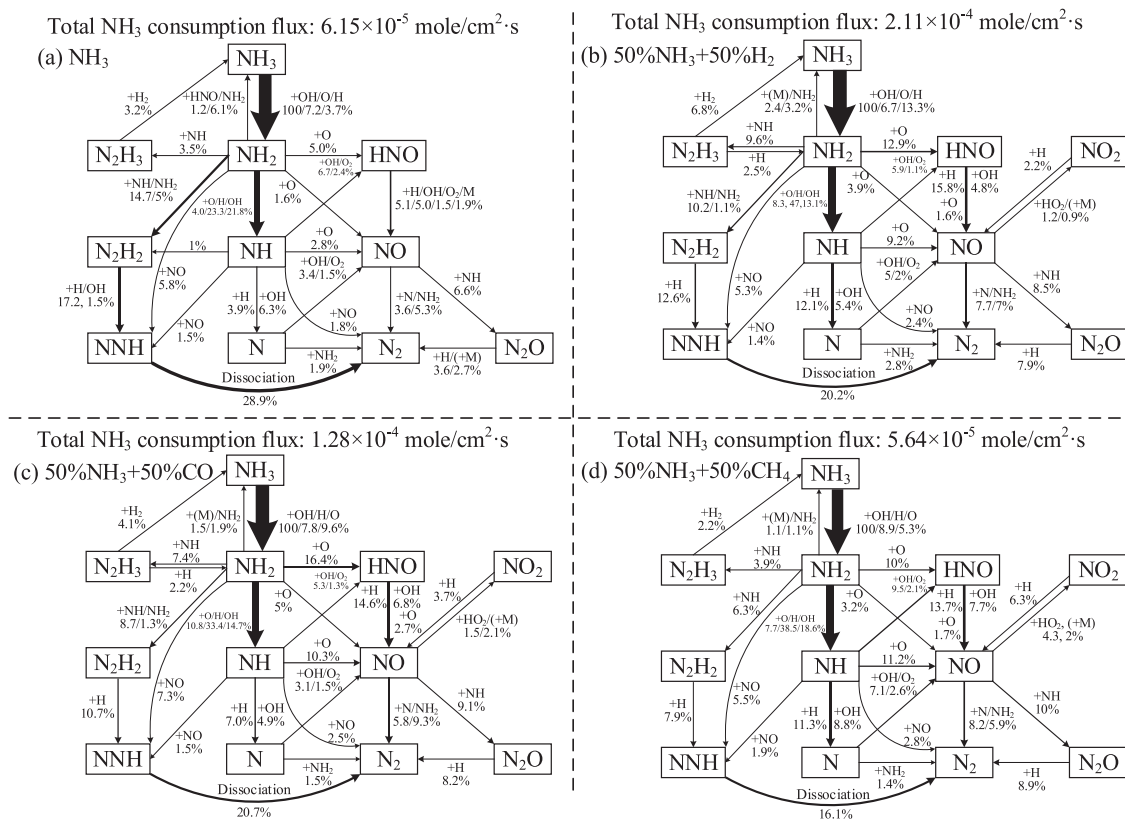


Fig. 15. Normalized consumption flux analysis of (a) NH_3/air , (b) $\text{NH}_3/\text{H}_2/\text{air}$, (c) $\text{NH}_3/\text{CO}/\text{air}$, and (d) $\text{NH}_3/\text{CH}_4/\text{air}$ flame at $\alpha = 0.5$, $\phi = 1.0$, and $T_i = 298$ K using the present model.

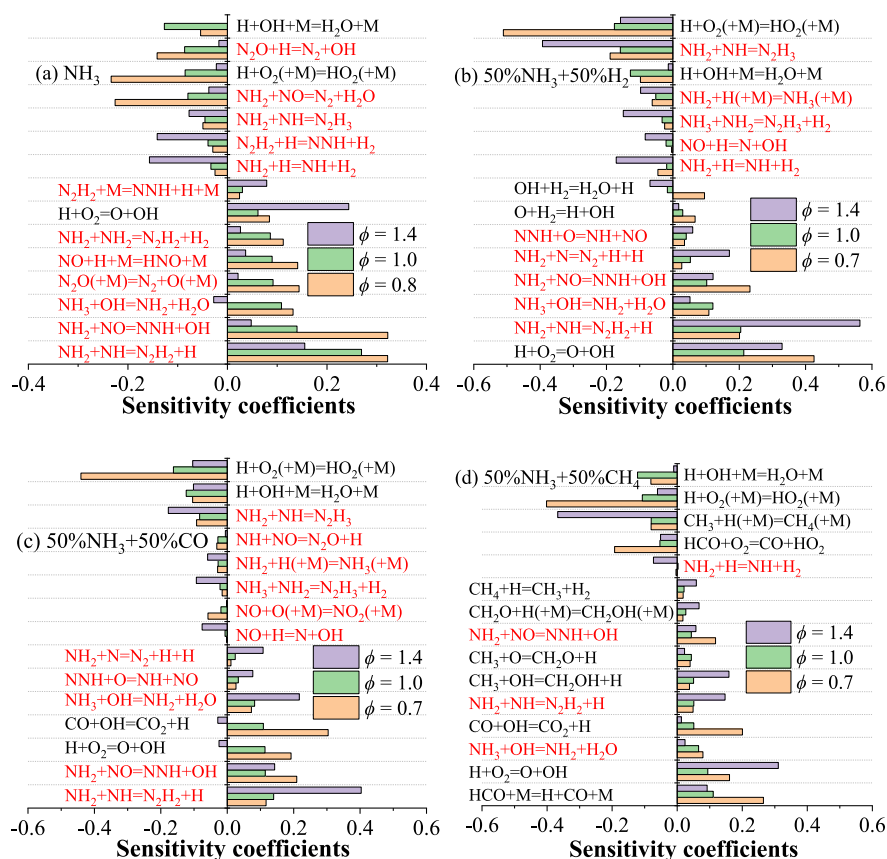


Fig. 16. Sensitivity analysis for S_L of (a) NH_3/air , (b) $\text{NH}_3/\text{H}_2/\text{air}$, (c) $\text{NH}_3/\text{CO}/\text{air}$, and (d) $\text{NH}_3/\text{CH}_4/\text{air}$ flame with various ϕ at $T_i = 423$ K.

at $\alpha > 0.4$. (3) NO_x chemistry becomes active as a result of the promoted reactions of NH_i with active radicals. Besides, it is noteworthy that NO_2 chemistry in the $\text{NH}_3/\text{CH}_4/\text{air}$ flame is more active than that in the $\text{NH}_3/\text{H}_2/\text{air}$ and $\text{NH}_3/\text{CO}/\text{air}$ flames. In the oxidation process of CH_4 , the HO_2 production not only depends on $\text{H} + \text{O}_2 (+\text{M}) = \text{HO}_2 (+\text{M})$, it also relies on the two elementary reactions of $\text{CH}_2\text{OH} + \text{O}_2 = \text{CH}_2\text{O} + \text{HO}_2$ and $\text{HCO} + \text{O}_2 = \text{CO} + \text{HO}_2$. HO_2 promotes the conversion of NO to NO_2 through $\text{NO} + \text{HO}_2 = \text{NO}_2 + \text{OH}$ [62]. Finally, it also can be found that the consumption flux of $\text{NH}_3/\text{CH}_4/\text{air}$ flame is the lowest among the four flames. The complicated oxidation process of CH_4 requires more active radicals, which inevitably inhibits the production of $\text{H}/\text{O}/\text{OH}$ radicals. Therefore, the consumption flux of NH_3 in the $\text{NH}_3/\text{CH}_4/\text{air}$ flame is significantly lower than that of $\text{NH}_3/\text{H}_2/\text{air}$ and $\text{NH}_3/\text{CO}/\text{air}$ flames. Since only half of the fuel is NH_3 , the consumption flux of NH_3 in the $\text{NH}_3/\text{CH}_4/\text{air}$ flame is even lower than that in the NH_3/air flame.

3.3.2. Effect of equivalence ratio (ϕ)

According to our previous study [8], the initial temperature has less effect on the sensitivity coefficients. To simulate the sensitivity coefficients of NH_3/air lean flame, the 15 most sensitive elementary reactions to S_L were selected for sensitivity analysis at $T_i = 423$ K shown in Fig. 16. Elementary reactions related to N chemistry are also marked in red.

In the stoichiometric flame of NH_3/air , $\text{NH}_2 + \text{NH} = \text{N}_2\text{H}_2 + \text{H}$ has the greatest positive effect on the increase of S_L , and $\text{H} + \text{OH} + \text{M} = \text{H}_2\text{O} + \text{M}$ has the greatest negative effect. In the lean flame of NH_3/air , $\text{NH}_2 + \text{NO} = \text{NNH} + \text{OH}$ has a similar sensitivity coefficient to that of $\text{NH}_2 + \text{NH} = \text{N}_2\text{H}_2 + \text{H}$ as the result of OH radicals production, demonstrating that the importance of NO chemistry significantly increases. It is worth mentioning that

$\text{NH}_2 + \text{NO} = \text{N}_2 + \text{H}_2\text{O}$, a chain-propagation reaction sharing the same reactants with $\text{NH}_2 + \text{NO} = \text{NNH} + \text{OH}$, has a negative sensitivity coefficient almost equivalent to that of $\text{H} + \text{O}_2 (+\text{M}) = \text{HO}_2 (+\text{M})$, implying the importance of NO reduction reactions for S_L of lean flames. In the rich flame of NH_3/air , there is not enough oxygen to react with abundant H and NH_i radicals to complete the oxidation, so the influence of the reaction of NH_2 with H radicals and the combination reaction of NH_i is enhanced. Since H radicals play different roles in the two reactions of $\text{NH}_2 + \text{H} = \text{NH} + \text{H}_2$ and $\text{NH}_2 + \text{NH} = \text{N}_2\text{H}_2 + \text{H}$, the two reactions have opposite effects on S_L .

In Fig. 16b, the largest positive sensitivity coefficient of $\text{H} + \text{O}_2 = \text{O} + \text{OH}$ and the largest negative sensitivity coefficient of $\text{H} + \text{O}_2 (+\text{M}) = \text{HO}_2 (+\text{M})$ imply that the lean flame of $\text{NH}_3/\text{H}_2/\text{air}$ is almost dominated by H_2 chemistry. In the rich flame of $\text{NH}_3/\text{H}_2/\text{air}$, H_2 promotes the conversion of NH_3 to NH_2 and NH_2 to NH . The importance of $\text{NH}_2 + \text{NH} = \text{N}_2\text{H}_2 + \text{H}$ and $\text{NH}_2 + \text{NH} = \text{N}_2\text{H}_3$ is further improved, and the rich flame of $\text{NH}_3/\text{H}_2/\text{air}$ is almost dominated by the two reactions. A similar phenomenon is also observed in the rich flame of $\text{NH}_3/\text{CO}/\text{air}$. In Fig. 16d, the lean flame of $\text{NH}_3/\text{CH}_4/\text{air}$ is almost dominated by CO chemistry, followed by H_2 chemistry. NO chemistry only has a slight positive effect on S_L . Meanwhile, the rich flame of $\text{NH}_3/\text{CH}_4/\text{air}$ is not dominated by the two reactions of NH_2 with NH . Although $\text{H}_2/\text{CO}/\text{CH}_4$ can bring more active radicals into the radical pool, CH_4 and its intermediates need more active radicals in competition with NH_3 . It leads to the obvious inhibition of NH_i oxidation, and the rich flame of $\text{NH}_3/\text{CH}_4/\text{air}$ is almost dominated by H_2 and CH_4 chemistry.

In addition, a higher positive sensitivity coefficient of $\text{NH}_2 + \text{NO} = \text{NNH} + \text{OH}$ than $\text{NH}_2 + \text{NH} = \text{N}_2\text{H}_2 + \text{H}$ can be observed in the lean flames of $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$, and $\text{NH}_3/\text{CH}_4/\text{air}$,

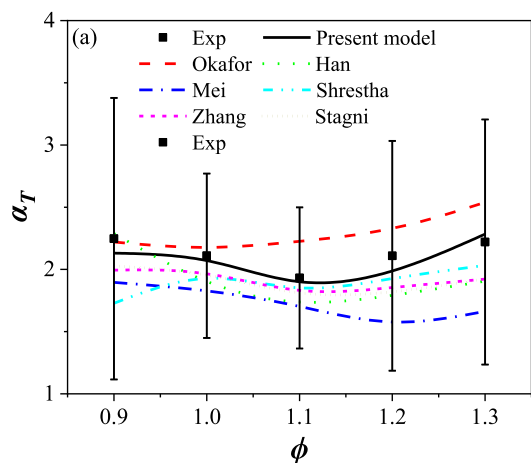


Fig. 17. Temperature dependence of different models ($\phi = 0.9\sim 1.3$, $T_i = 423$ K).

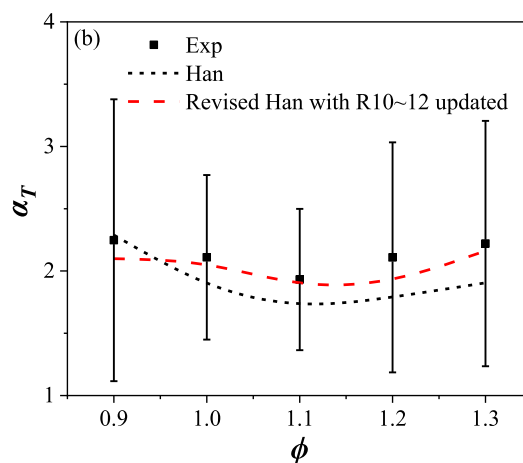


Fig. 18. Temperature dependence of Han model with and without R10~12 updated ($\phi = 0.9\sim 1.3$, $T_i = 423$ K).

implying that the NO chemistry in the lean flames of $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$, and $\text{NH}_3/\text{CH}_4/\text{air}$ is more active than that in the lean flame of NH_3/air . In the lean flame of NH_3/air , even though NH_2 is partially converted to NO through the pathway of $\text{NH}_2 \rightarrow \text{HNO} \rightarrow \text{NO}$, the remaining abundant NH_2 still reduces NO through the reactions of $\text{NH}_2 + \text{NO} = \text{N}_2 + \text{H}_2\text{O}$ and $\text{NH}_2 + \text{NO} = \text{NNH} + \text{OH}$. It is easily visible in Fig. 15 that the conversion of NH_2 radicals to NO in the stoichiometric flames of $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$, and $\text{NH}_3/\text{CH}_4/\text{air}$ is significantly improved, while the promotion of NO reduction by NH_2 radicals is very slight. It indicates that $\text{H}_2/\text{CO}/\text{CH}_4$ will promote the conversion of NH_2 to NO, and relatively inhibit the NO reduction by NH_2 , resulting in a possible increase in NO emissions. Experimental result consistent with the inferences has been found by Wargadalam et al. [55]. Therefore, substitution with highly reactive fuels may not be a suitable option for suppressing the NO production of NH_3 combustion.

3.3.3. Effect of initial temperature (T_i)

According to [41], the temperature dependence can be expressed as Eq. (1). Where $T_i^0 = 298$ K, S_L^0 is the laminar burning velocity at $T_i = 298$ K.

$$\frac{S_L}{S_L^0} = \left(\frac{T_i}{T_i^0} \right)^{\alpha_T} \quad (1)$$

Fig. 17 shows the temperature dependence of different models with $\phi = 0.9\sim 1.3$ at $T_i = 423$ K. Okafor model shows too high temperature dependence, while other models show too low temperature dependence. Only the temperature dependence of present model shows the best consistency with the temperature dependence of the experiment. As stated in Section 2.2, this phenomenon is closely related to the difference in ammonia chemistry, especially R10~12. The rate constants of the original R10 is independent of temperature, and the reaction rate of the updated R10 is positively related to T_i . In addition, it can be seen from Fig. 16a that no matter how the equivalence ratio changes, N_2H_2 chemistry plays an important role in promoting the laminar flame propagation of NH_3 . Through the update of R10~12, proper NNH chemistry and N_2H_2 chemistry can better reproduce the oxidation pathway of $\text{NH}_2 \rightarrow \text{N}_2\text{H}_2 \rightarrow \text{NNH} \rightarrow \text{N}_2$. Accordingly, we try to apply the updates of R10~12 to other models, e.g., Han model. As shown in Fig. 18, a better temperature dependence can be observed.

Fig. 19 shows the temperature dependence of NH_3/air , $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$, and $\text{NH}_3/\text{CH}_4/\text{air}$ flame. The slope in Fig. 19 represents the temperature coefficient (α_T), reflecting the temperature dependence. Limited by the measurement method, the S_L of 50% $\text{NH}_3/50\%$ H_2/air flame (50% $\text{NH}_3+50\%$ H_2) at elevated

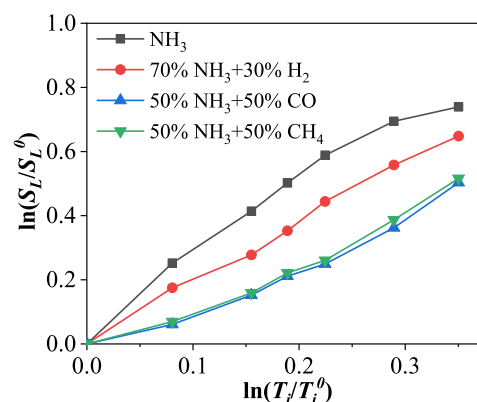


Fig. 19. Temperature dependence of NH_3/air , $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$, and $\text{NH}_3/\text{CH}_4/\text{air}$ flame.

temperatures almost cannot be measured. The simulated temperature dependence of $\text{NH}_3/\text{H}_2/\text{air}$ flame with different H_2 additions is shown in S6 of Supplementary Material, only a minor difference has been observed between the α_T of 50% $\text{NH}_3/50\%$ H_2/air and 70% $\text{NH}_3/30\%$ H_2/air flames. The strongest temperature dependence of NH_3/air flame is observed in Fig. 19, implying that preheating combustion technology is an effective method to improve the reactivity of NH_3/air flame. α_T of $\text{NH}_3/\text{H}_2/\text{air}$ flame is second only to that of NH_3/air flame. In the past decades, NH_3 was first used as a flame inhibitor for H_2 [63]. Therefore, preheating combustion technology is usually only used with low addition of H_2 . α_T of $\text{NH}_3/\text{CO}/\text{air}$ and $\text{NH}_3/\text{CH}_4/\text{air}$ flames are only 0.5 and 0.52, respectively. Since the conversion of CH_4 to CO_2 depends on CO as the intermediate and oxidation product, and CO chemistry almost dominates the lean and stoichiometric flame of $\text{NH}_3/\text{CH}_4/\text{air}$ as shown in Fig. 16d, a similar temperature dependence is found.

4. Conclusion

In this work, an experimental and kinetic modeling study on the premixed laminar flame propagation of NH_3/air , $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$, and $\text{NH}_3/\text{CH}_4/\text{air}$ mixtures was performed in a constant-volume combustion vessel. The experimental conditions cover various $\text{H}_2/\text{CO}/\text{CH}_4$ mole fractions in fuel (α) in the mixtures, equivalence ratios (ϕ) of $0.7\sim 1.4$, and initial temperature (T_i) of $298\sim 423$ K. A detailed kinetic model was developed and validated based on the experimental and literature data. The conclusions drawn through experiment and kinetic analysis are as follows:

- (1) The S_L of $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$, and $\text{NH}_3/\text{CH}_4/\text{air}$ flames has a different dependence on α , being almost exponentially for $\text{NH}_3/\text{H}_2/\text{air}$ flame, non-linear for $\text{NH}_3/\text{CO}/\text{air}$ flame, and nearly linear for $\text{NH}_3/\text{CH}_4/\text{air}$ flame.
- (2) A kinetic model with 169 species and 1268 elementary reactions was developed and validated against the S_L , NOx emissions, NH_3/NO interaction, and ignition delay times measured by this work and literature. In addition to the good fuel flexibility, the S_L with various ϕ and T_i also can be predicted accurately by the updated NH_2 , NNH , and N_2H_2 chemistries. The temperature dependence of NH_3/air flame and the predictions of N_2O formation respectively, is improved with the updated NNH and NOx chemistries.
- (3) H_2 , CO , and CH_4 can promote the production of active radicals ($\text{H}/\text{O}/\text{OH}$). H radicals play a more important role in the $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$, and $\text{NH}_3/\text{CH}_4/\text{air}$ flames than in the NH_3/air flame. Besides, the combination reaction of NH_2 and NH are inhibited, while the formation and reduction of NOx become more active by the pathways of NH_2 with O and NO as a result of the increased concentrations of NH_i and active radicals.
- (4) In the lean flames of $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$, and $\text{NH}_3/\text{CH}_4/\text{air}$, H_2 , CO , and CH_4 promotes the conversion of NH_2 to NO , and relatively inhibit the NO reduction by NH_2 , resulting in a possible increase in NO emissions. In the rich flames of $\text{NH}_3/\text{H}_2/\text{air}$ and $\text{NH}_3/\text{CO}/\text{air}$, NH_2 and NH produce N_2H_2 and N_2H_3 , which dominate the promotion and inhibition of laminar flame velocity propagation, respectively. Meanwhile, the rich flame of $\text{NH}_3/\text{CH}_4/\text{air}$ is almost dominated by H_2 and CH_4 chemistry for the fewer production of NH_i radicals.
- (5) Although H_2 , CO , and CH_4 can increase the S_L , it harms the temperature dependence of NH_3/air flame. Due to the importance of CO chemistry in CH_4 oxidation, the temperature dependence of $\text{NH}_3/\text{CO}/\text{air}$ and $\text{NH}_3/\text{CH}_4/\text{air}$ flames are similar. The highest temperature dependence of NH_3/air flame indicates that the air preheating technology has greater potential in improving the reactivity of NH_3/air flame.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.combustflame.2022.112536](https://doi.org/10.1016/j.combustflame.2022.112536).

References

- [1] H. Kobayashi, A. Hayakawa, K.D.K.A. Somaratne, E.C. Okafor, Science and technology of ammonia combustion, *Proc. Combust. Inst.* 37 (2019) 109–133.
- [2] A. Valera-Medina, F. Amer-Hatem, A.K. Azad, I.C. Dedoussi, M. de Joannon, R.X. Fernandes, P. Glarborg, H. Hashemi, X. He, S. Mashruk, J. McGowan, C. Mounaim-Rousselle, A. Ortiz-Prado, A. Ortiz-Valera, I. Rossetti, B. Shu, M. Yehia, H. Xiao, M. Costa, Review on ammonia as a potential fuel: from synthesis to economics, *Energy Fuels* 35 (2021) 6964–7029.
- [3] A. Valera-Medina, H. Xiao, M. Owen-Jones, W.I.F. David, P.J. Bowen, Ammonia for power, *Prog. Energy Combust. Sci.* 69 (2018) 63–102.
- [4] K.P. Shrestha, C. Lhuillier, A.A. Barbosa, P. Brequigny, F. Contino, C. Mounaim-Rousselle, L. Seidel, F. Mauss, An experimental and modeling study of ammonia with enriched oxygen content and ammonia/hydrogen laminar flame speed at elevated pressure and temperature, *Proc. Combust. Inst.* 38 (2020) 2163–2174.
- [5] X. Han, Z. Wang, M. Costa, Z. Sun, Y. He, K. Cen, Experimental and kinetic modeling study of laminar burning velocities of NH_3/air , $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$ and $\text{NH}_3/\text{CH}_4/\text{air}$ premixed flames, *Combust. Flame* 206 (2019) 214–226.
- [6] X. Han, Z. Wang, Y. He, Y. Zhu, K. Cen, Experimental and kinetic modeling study of laminar burning velocities of $\text{NH}_3/\text{syngas}/\text{air}$ premixed flames, *Combust. Flame* 213 (2020) 1–13.
- [7] B. Mei, S. Ma, Y. Zhang, X. Zhang, W. Li, Y. Li, Exploration on laminar flame propagation of ammonia and syngas mixtures up to 10 atm, *Combust. Flame* 220 (2020) 368–377.
- [8] S. Zhou, W. Yang, H. Tan, Q. An, J. Wang, H. Dai, X. Wang, X. Wang, S. Deng, Experimental and kinetic modeling study on $\text{NH}_3/\text{syngas}/\text{air}$ and $\text{NH}_3/\text{bio-syngas}/\text{air}$ premixed laminar flames at elevated temperature, *Combust. Flame* 233 (2021) 111594.
- [9] B. Mei, X. Zhang, S. Ma, M. Cui, H. Guo, Z. Cao, Y. Li, Experimental and kinetic modeling investigation on the laminar flame propagation of ammonia under oxygen enrichment and elevated pressure conditions, *Combust. Flame* 210 (2019) 236–246.
- [10] Z. Tian, Y. Li, L. Zhang, P. Glarborg, F. Qi, An experimental and kinetic modeling study of premixed $\text{NH}_3/\text{CH}_4/\text{O}_2/\text{Ar}$ flames at low pressure, *Combust. Flame* 156 (2009) 1413–1426.
- [11] E.C. Okafor, Y. Naito, S. Colson, A. Ichikawa, T. Kudo, A. Hayakawa, H. Kobayashi, Experimental and numerical study of the laminar burning velocity of $\text{CH}_4\text{--NH}_3\text{--air}$ premixed flames, *Combust. Flame* 187 (2018) 185–198.
- [12] G.P. Smith, D.M. Golden, M. Frenklach, N.W. Moriarty, B. Eiteneer, M. Goldenberg, GRI Mech 3.0. Gas Research Institute, available at http://www.me.berkeley.edu/gri_mech/.
- [13] K.P. Shrestha, L. Seidel, T. Zeuch, F. Mauss, Detailed kinetic mechanism for the oxidation of ammonia including the formation and reduction of nitrogen oxides, *Energy Fuels* 32 (2018) 10202–10217.
- [14] A. Stagni, C. Cavallotti, S. Arunthanayothin, Y. Song, O. Herbinet, F. Battin-Leclerc, T. Faravelli, An experimental, theoretical and kinetic-modeling study of the gas-phase oxidation of ammonia, *React. Chem. Eng.* 5 (2020) 696–711.
- [15] S. Arunthanayothin, A. Stagni, Y. Song, O. Herbinet, T. Faravelli, F. Battin-Leclerc, Ammonia-methane interaction in jet-stirred and flow reactors: an experimental and kinetic modeling study, *Proc. Combust. Inst.* 38 (2021) 345–353.
- [16] W.K. Metcalfe, S.M. Burke, S.S. Ahmed, H.J. Curran, A hierarchical and comparative kinetic modeling study of C1–C2 hydrocarbon and oxygenated fuels, *Int. J. Chem. Kinet.* 45 (2013) 638–675.
- [17] X. Zhang, S.P. Moosakutty, R.P. Rajan, M. Younes, S.M. Sarathy, Combustion chemistry of ammonia/hydrogen mixtures: Jet-stirred reactor measurements and comprehensive kinetic modeling, *Combust. Flame* 234 (2021) 111653.
- [18] Y. Li, S.M. Sarathy, Probing hydrogen-nitrogen chemistry: a theoretical study of important reactions in N_2H_2 , HCN and HNCO oxidation, *Int. J. Hydrogen Energy* 45 (2020) 23624–23637.
- [19] R. Tang, Q. Xu, J. Pan, J. Gao, Z. Wang, H. Wei, G. Shu, An experimental and modeling study of ammonia oxidation in a jet stirred reactor, *Combust. Flame* 240 (2022) 112007.
- [20] A.P. Kelley, C.K. Law, Nonlinear effects in the extraction of laminar flame speeds from expanding spherical flames, *Combust. Flame* 156 (2009) 1844–1851.
- [21] O. Mathieu, E.L. Petersen, Experimental and modeling study on the high-temperature oxidation of ammonia and related NOx chemistry, *Combust. Flame* 162 (2015) 554–570.
- [22] G.T. Linteris, M.D. Rumminger, V.I. Babushok, Catalytic inhibition of laminar flames by transition metal compounds, *Prog. Energy Combust. Sci.* 34 (2008) 288–329.
- [23] C.-W. Zhou, Y. Li, U. Burke, C. Banyon, K.P. Somers, S. Ding, S. Khan, J.W. Hargis, T. Sikes, O. Mathieu, E.L. Petersen, M. Alabbad, A. Farooq, Y. Pan, Y. Zhang, Z. Huang, J. Lopez, Z. Loparo, S.S. Vasu, H.J. Curran, An experimental and chemical kinetic modeling study of 1,3-butadiene combustion: ignition delay time and laminar flame speed measurements, *Combust. Flame* 197 (2018) 423–438.
- [24] S.J. Klippenstein, P. Glarborg, Theoretical kinetics predictions for $\text{NH}_2 + \text{HO}_2$, *Combust. Flame* 236 (2022) 111787.
- [25] J.C. Mackie, G.B. Bacskay, Quantum chemical study of the mechanism of reaction between $\text{NH} (X^3\Sigma^-)$ and H_2 , H_2O , and CO_2 under combustion conditions, *J. Phys. Chem.* 109 (2005) 11967–11974.
- [26] D.P. Linder, X. Duan, M. Page, Ab initio variational transition state theory calculations for the $\text{H} + \text{NH}_2$ tautomerization, $\text{H}_2 + \text{NH}$ hydrogen abstraction reaction on the triplet potential energy surface, *J. Phys. Chem.* 99 (1995) 11458–11463.
- [27] S.J. Klippenstein, L.B. Harding, P. Glarborg, J.A. Miller, The role of NNH in NO formation and control, *Combust. Flame* 158 (2011) 774–789.
- [28] A.M. Dean, J.W. Bozzelli, Combustion chemistry of nitrogen, *Gas-Phase, Combust. Chem.* (2000) 125–341.
- [29] N. Cohen, K.R. Westberg, Chemical kinetic data sheets for high-temperature reactions. Part II, *J. Phys. Chem. Ref. Data* 20 (1991) 1211–1311.
- [30] P. Glarborg, J.A. Miller, B. Ruscic, S.J. Klippenstein, Modeling nitrogen chemistry in combustion, *Prog. Energy Combust. Sci.* 67 (2018) 31–68.

- [31] J.E. Chavarrio Cañas, M. Monge-Palacios, X. Zhang, S.M. Sarathy, Probing the gas-phase oxidation of ammonia: addressing uncertainties with theoretical calculations, *Combust. Flame* 235 (2022) 111708.
- [32] D.L. Baulch, C.T. Bowman, C.J. Cobos, R.A. Cox, T. Just, J.A. Kerr, M.J. Pilling, D. Stocker, J. Troe, W. Tsang, R.W. Walker, J. Warnatz, Evaluated kinetic data for combustion modeling: supplement II, *J. Phys. Chem. Ref. Data* 34 (2005) 757–1397.
- [33] J.A. Miller, C.T. Bowman, Mechanism and modeling of nitrogen chemistry in combustion, *Prog. Energy Combust. Sci.* 15 (1989) 287–338.
- [34] Chemkin-Pro 15151, Reaction Design, San Diego, (2009).
- [35] Y. Song, H. Hashemi, J.M. Christensen, C. Zou, P. Marshall, P. Glarborg, Ammonia oxidation at high pressure and intermediate temperatures, *Fuel* 181 (2016) 358–365.
- [36] A. Hayakawa, T. Goto, R. Mimoto, Y. Arakawa, T. Kudo, H. Kobayashi, Laminar burning velocity and Markstein length of ammonia/air premixed flames at various pressures, *Fuel* 159 (2015) 98–106.
- [37] U.J. Pfahl, M.C. Ross, J.E. Shepherd, Flammability limits, ignition energy, and flame speeds in H_2 - CH_4 - NH_3 - N_2 - O_2 - N_2 mixtures, *Combust. Flame* 123 (2000) 140–158.
- [38] P.D. Ronney, H.Y. Wachman, Effect of gravity on laminar premixed gas combustion I: flammability limits and burning velocities, *Combust. Flame* 62 (1985) 107–119.
- [39] K. Takizawa, T. Akifumi, T. Kazuaki, K. Shigeo, S. Akira, Burning velocity measurements of nitrogen-containing compounds, *J. Hazard. Mater.* 155 (2008) 144–152.
- [40] V.F. Zakaznov, L.A. Kursheva, Z.I. Fedina, Determination of normal flame velocity and critical diameter of flame extinction in ammonia-air mixture, *Combust. Explo. Shock* 14 (1978) 710–713.
- [41] X. Han, Z. Wang, Y. He, Y. Liu, Y. Zhu, A.A. Konnov, The temperature dependence of the laminar burning velocity and superadiabatic flame temperature phenomenon for NH_3 /air flames, *Combust. Flame* 217 (2020) 314–320.
- [42] J.H. Lee, S.I. Lee, O.C. Kwon, Effects of ammonia substitution on hydrogen/air flame propagation and emissions, *Int. J. Hydrogen Energy* 35 (2010) 11332–11341.
- [43] A. Ichikawa, A. Hayakawa, Y. Kitagawa, S.K.D.K. Amila, T. Kudo, H. Kobayashi, Laminar burning velocity and Markstein length of ammonia/hydrogen/air premixed flames at elevated pressures, *Int. J. Hydrogen Energy* 40 (2015) 9570–9578.
- [44] C. Lhuillier, P. Brequigny, N. Lamoureux, F. Contino, C. Mounaïm-Rousselle, Experimental investigation on laminar burning velocities of ammonia/hydrogen/air mixtures at elevated temperatures, *Fuel* 263 (2020) 116653.
- [45] P. Kumar, T.R. Meyer, Experimental and modeling study of chemical-kinetics mechanisms for H_2 - NH_3 -air mixtures in laminar premixed jet flames, *Fuel* 108 (2013) 166–176.
- [46] U. mechanism, Chemical-kinetic Mechanisms For Combustion applications, Mechanical and Aerospace Engineering (Combustion Research), University of California at San Diego, 2018 <http://web.eng.ucsd.edu/mae/groups/combustion/mechanism.html>.
- [47] J. Otomo, M. Koshi, T. Mitsumori, H. Iwasaki, K. Yamada, Chemical kinetic modeling of ammonia oxidation with improved reaction mechanism for ammonia/air and ammonia/hydrogen/air combustion, *Int. J. Hydrogen Energy* 43 (2018) 3004–3014.
- [48] S. Wang, Z. Wang, A.M. Elbaz, X. Han, Y. He, M. Costa, A.A. Konnov, W.L. Roberts, Experimental study and kinetic analysis of the laminar burning velocity of NH_3 /syngas/air, NH_3 /CO/air and NH_3 / H_2 /air premixed flames at elevated pressures, *Combust. Flame* 221 (2020) 270–287.
- [49] E.C. Okafor, Y. Nagano, T. Kitagawa, Experimental and theoretical analysis of cellular instability in lean H_2 - CH_4 -air flames at elevated pressures, *Int. J. Hydrogen Energy* 41 (2016) 6581–6592.
- [50] M.L. Rightley, F.A. Williams, Burning velocities of CO flames, *Combust. Flame* 110 (1997) 285–297.
- [51] T. Shu, Y. Xue, Z. Zhou, Z. Ren, An experimental study of laminar ammonia/methane/air premixed flames using expanding spherical flames, *Fuel* 290 (2021) 120003.
- [52] L. Dai, S. Gersen, P. Glarborg, A. Mokhov, H. Levinsky, Autoignition studies of NH_3 / CH_4 mixtures at high pressure, *Combust. Flame* 218 (2020) 19–26.
- [53] E.C. Okafor, Y. Naito, S. Colson, A. Ichikawa, T. Kudo, A. Hayakawa, H. Kobayashi, Measurement and modelling of the laminar burning velocity of methane-ammonia-air flames at high pressures using a reduced reaction mechanism, *Combust. Flame* 204 (2019) 162–175.
- [54] P. Dagaut, A. Nicolle, Experimental and kinetic modeling study of the effect of SO_2 on the reduction of NO by ammonia, *Proc. Combust. Inst.* 30 (2005) 1211–1218.
- [55] V.J. Wargadalam, G. Löffler, F. Winter, H. Hofbauer, Homogeneous formation of NO and N_2O from the oxidation of HCN and NH_3 at 600–1000 °C, *Combust. Flame* 120 (2000) 465–478.
- [56] M.U. Alzueta, L. Ara, V.D. Mercader, M. Delogu, R. Bilbao, Interaction of NH_3 and NO under combustion conditions. Experimental flow reactor study and kinetic modeling simulation, *Combust. Flame* 235 (2022) 111691.
- [57] O. Mathieu, M.M. Kopp, E.L. Petersen, Shock-tube study of the ignition of multi-component syngas mixtures with and without ammonia impurities, *Proc. Combust. Inst.* 34 (2013) 3211–3218.
- [58] F. Kasuya, P. Glarborg, J.E. Johnsson, K. Dam-Johansen, The thermal DeNOx process: influence of partial pressures and temperature, *Chem. Eng. Sci.* 50 (1995) 1455–1466.
- [59] B. Shu, S.K. Vallabhuni, X. He, G. Issayev, K. Moshhammer, A. Farooq, R.X. Fernandes, A shock tube and modeling study on the autoignition properties of ammonia at intermediate temperatures, *Proc. Combust. Inst.* 37 (2019) 205–211.
- [60] Z. Sun, Y. Deng, S. Song, J. Yang, W. Yuan, F. Qi, Experimental and kinetic modeling study of the homogeneous chemistry of NH_3 and NOx with CH_4 at the diluted conditions, *Combust. Flame* 112015 (2022) 112015, doi:10.1016/j.combustflame.2022.
- [61] Z. Wang, S. Wang, R. Whiddon, X. Han, Y. He, K. Cen, Effect of hydrogen addition on laminar burning velocity of CH_4 /DME mixtures by heat flux method and kinetic modeling, *Fuel* 232 (2018) 729–742.
- [62] R. Li, A.A. Konnov, G. He, F. Qin, D. Zhang, Chemical mechanism development and reduction for combustion of NH_3 / H_2 / CH_4 mixtures, *Fuel* 257 (2019) 116059.
- [63] D.H. Um, J.M. Joo, S. Lee, O.C. Kwon, Combustion stability limits and NOx emissions of nonpremixed ammonia-substituted hydrogen-air flames, *Int. J. Hydrogen Energy* 38 (2013) 14854–14865.